Experiment- 9:

OBJECTIVE:

To formulate a continuous-time LTI system from a differential equation, obtain its transfer function under zero initial conditions, simulate impulse and step responses, and relate the system poles to its transient behavior and stability.

Experimental Procedure:

1. Write the governing first-order differential equation and identify its coefficients.

2. Apply the Laplace transform assuming zero initial conditions.

3. Form H(s) = 1/(s + 2) and enter numerator and denominator coefficients [1] and [1 2] in MATLAB.

4. Obtain the pole and zero of the system; verify that the pole is at s = −2.

5. Plot the impulse and step responses.

6. Apply a test input using lsim and observe the system response.

7. Relate the response shape and settling behavior to the pole at s = −2.

Code:

```
clc;

clear;

close all;


% Continuous-Time LTI System

% dy(t)/dt + 2y(t) = x(t)


% Transfer Function H(s) = 1/(s+2)

num = [1];

den = [1 2];


sys = tf(num,den);


% Display system information
```

```
disp('Transfer Function:');
sys

disp('Poles:');
disp(pole(sys));

disp('Zeros:');
disp(zero(sys));

% Time vector
t = 0:0.01:5;

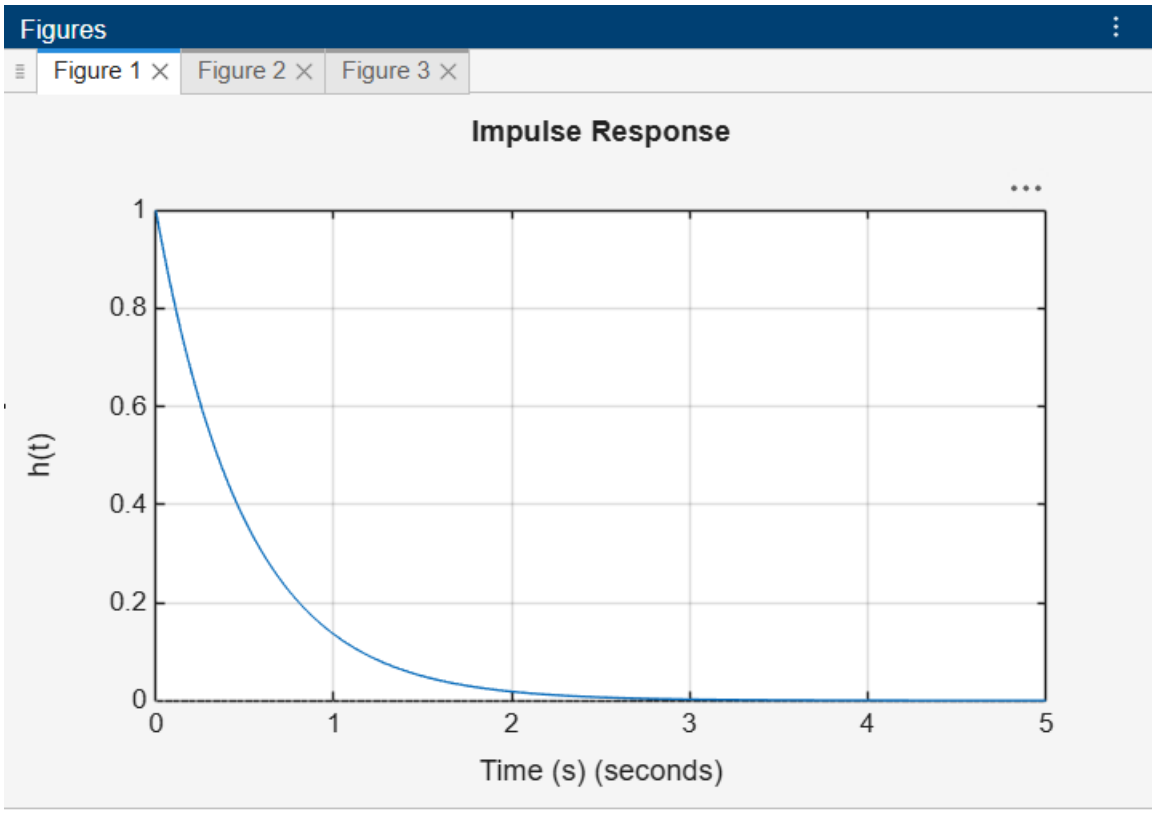
% Impulse Response
figure;
impulse(sys,t);
grid on;
title('Impulse Response');
xlabel('Time (s)');
ylabel('h(t)');

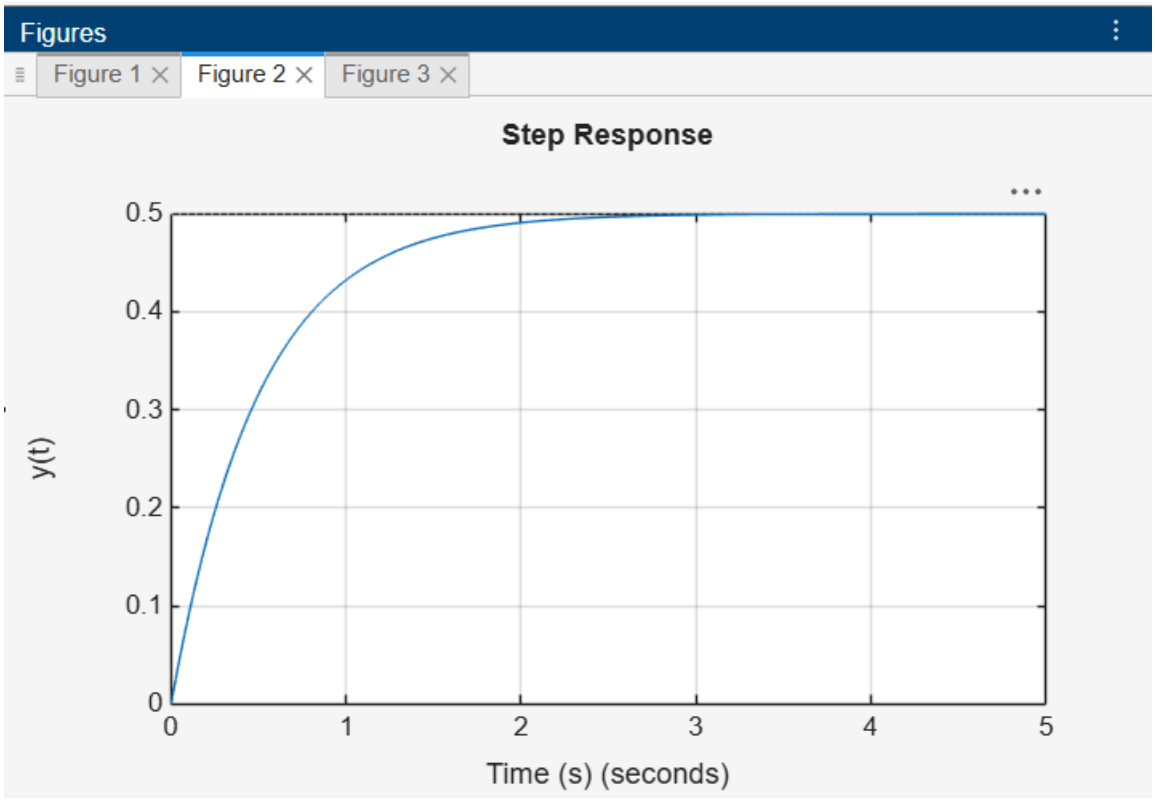
% Step Response
figure;
step(sys,t);
grid on;
title('Step Response');
xlabel('Time (s)');
ylabel('y(t)');
```

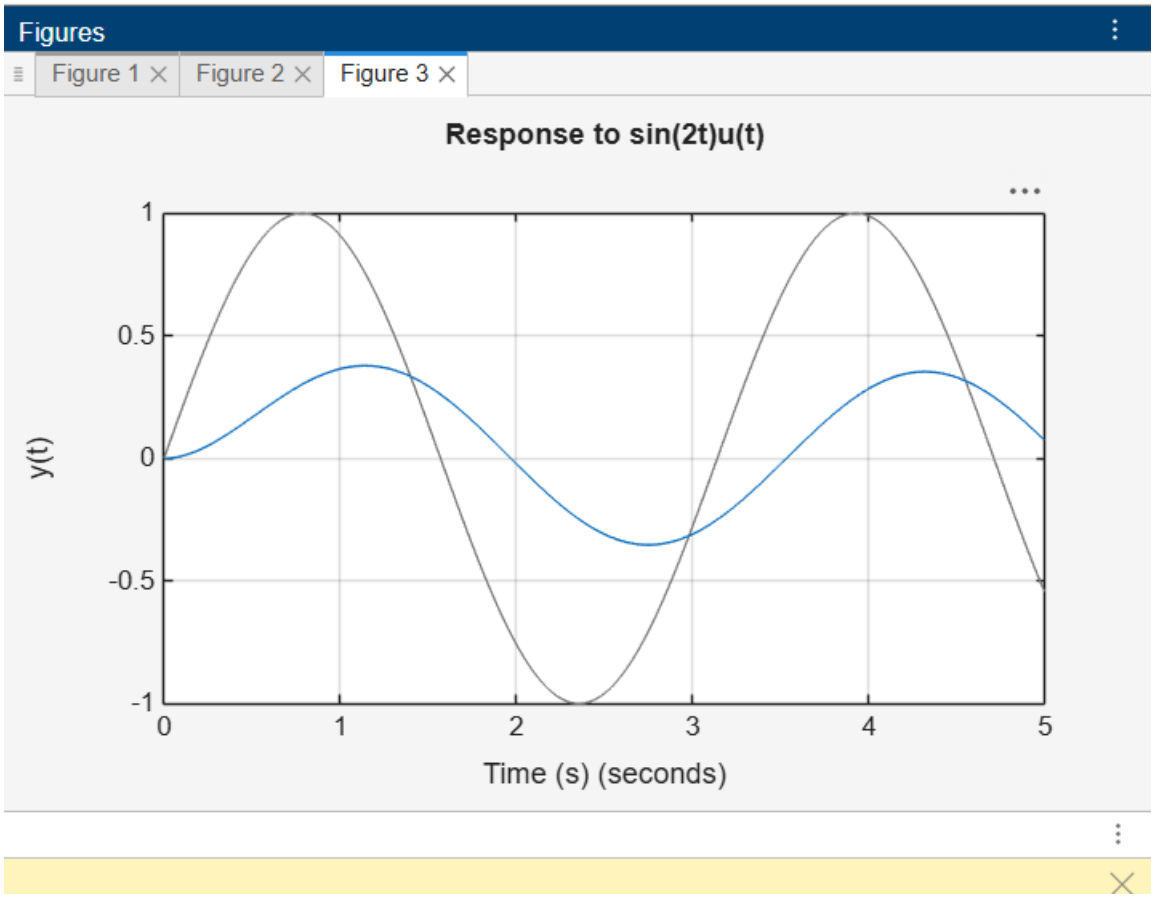
```
% Response to sinusoidal input
x = sin(2*t);

figure;
lsim(sys,x,t);
grid on;
title('Response to sin(2t)u(t)');
xlabel('Time (s)');
ylabel('y(t)');
```

output:




Figures
Figure 1
Figure 2
Figure 3
Step Response
0.5
0.4
0.3
0.2
0.1
0
y(t)
0
1
2
3
4
5
Time (s) (seconds)


Figures
Figure 1
Figure 2
Figure 3
Response to sin(2t)u(t)
1
0.5
0
-0.5
-1
y(t)
0
1
2
3
4
5
Time (s) (seconds)

Result:

Thus, the continuous-time LTI system was successfully formulated from the given differential equation. The transfer function was obtained under zero initial conditions, and the impulse and step responses were simulated using MATLAB. The system pole was found at \(s=-2\), indicating that the system is stable and its transient response decays exponentially with time.

Here is the Laboratory Performance Assessment Rubric in table format, similar to the picture you provided.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Evaluation Criterion	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**
Differential equation formulation	Correctly interprets the given differential equation, identifies system variables and coefficients, and formulates the LTI model independently.	Formulates the model correctly with minor notation or coefficient errors.	Identifies the main variables but needs guidance or makes noticeable formulation errors.	Unable to interpret the equation or formulate a suitable LTI model.	20
Transfer function and state-space representation	Correctly obtains the transfer function under zero initial conditions and derives a valid state-space	Obtains a valid representation with minor algebraic or matrix-entry errors.	Produces a partial model but has errors or incomplete representation.	Unable to derive a valid transfer function or state-space model.	20

	representation.				
MATLAB implementation and response simulation	Implements the model independently in MATLAB and correctly simulates the specified input and system output.	Simulation is substantially correct with minor coding or setup errors.	Runs a partial simulation with assistance or noticeable coding errors.	Unable to implement or simulate the system correctly.	20
Response analysis and dynamic interpretation	Accurately interprets transient and steady-state behaviour, relates the response to system coefficients, and checks whether the output is bounded.	Interprets the response correctly with minor gaps in explanation.	Provides a basic interpretation, but transient/steady-state analysis is incomplete.	Unable to interpret the response or relate it to system behaviour.	20
Plots, verification, record and conclusion	Presents clearly labelled input/output plots, verifies expected behaviour, maintains a complete record, and states a	Plots and record are clear with minor omissions or limited verification.	Plots, verification, record or conclusion are incomplete and require improvement.	Missing or unclear plots, verification, record or conclusion.	20

	technically justified conclusion.				
Total Weightage					100